Forecasting Crude Oil Prices Using LMD, ARIMA, and XGBoost

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Introduction

- Crude oil is a crucial global resource affecting environment, politics, and the economy
- Oil prices are highly volatile and influenced by multiple factors
  - Market dynamics
  - Geopolitical events
  - External shocks
- Price movements are random, volatile, and disorderly
  - Non-stationarity
  - Non-linearity
- Single models fail to capture all complexities
  - Noise
  - Non-linearity
  - Non-stationarity
- **Solution: Hybrid Model**
  - Local Mean Decomposition (LMD)
  - ARIMA
  - XGBoost

What the Paper Does

- The paper proposes a hybrid forecasting model for crude oil prices
- Key idea:
  - Combine statistical and machine learning methods
  - Capture both linear and non-linear patterns
- Previous approaches:
  - Traditional models (ARIMA, GARCH)
    $\rightarrow$ Fail to capture non-linearity
  - Machine learning models (ANN, SVR, Random Forest)
    $\rightarrow$ Prone to overfitting
- Motivation:
  - Non-linearity
  - Non-stationarity
  - Noise
- **Hence, a hybrid approach is required**

Proposed Hybrid Approach

- Step 1: Decompose data using LMD
  - Separates data into components based on frequency
- Step 2: Separate components
  - Deterministic components
  - Stochastic components
- Step 3: Apply ARIMA
  - Models linear time-series structure
- Step 4: Residual Analysis
  - Residuals contain nonlinear patterns
- Step 5: Apply XGBoost
  - Captures nonlinear residual patterns
- **Final Output: Improved forecasting accuracy**

Statistical Overview of Data

Dataset Used

- West Texas Intermediate (WTI) crude oil prices
- Source: U.S. Energy Information Administration (EIA)

Time Period

- March 2018 – February 2022

Data Split

- Training Set: 75%
- Testing Set: 25%

Key Observations

- Data shows strong volatility and fluctuations
- Non-stationary behavior (confirmed by ADF and PP tests)
- Presence of trend and structural changes over time

Local Mean Decomposition (LMD)

Step 1: Compute Local Mean and Envelope

$$m_i = \frac{o_i + o_{i+1}}{2}, \quad \alpha_i = \frac{|o_i - o_{i+1}|}{2}$$

Step 2: Smooth Mean and Envelope

$$m_{11}(t), \quad \alpha_{11}(t)$$

Step 3: Remove Local Mean

$$h(t) = x(t) - m_{11}(t)$$

LMD (Contd.)

Step 4: Amplitude Demodulation

$$s_{11}(t) = \frac{h(t)}{\alpha_{11}(t)}$$

Step 5: Construct Envelope and Product Function

$$\alpha_1(t) = \prod_{q=1}^{o} \alpha_{1q}(t)$$

$$PF_1(t) = \alpha_1(t)\, s_{1o}(t)$$

Step 6: Final Decomposition

$$x(t) = \sum_{p=1}^{k} PF_p(t) + u_k(t)$$

Deterministic vs Stochastic (1/2)

- After LMD, the series is decomposed into Product Functions (PFs)
- LMD does not classify components
- Classification uses **Average Mutual Information (AMI)**
  - Measures predictability / dependence
- **Identification:**
  - Low AMI $+$ irregular $\rightarrow$ Stochastic
  - High AMI $+$ structured $\rightarrow$ Deterministic
  - Supported by visual inspection

Deterministic vs Stochastic (2/2)

- **In this paper:**
  - PF1 $\rightarrow$ Stochastic (high-frequency, noisy)
  - PF2, PF3, PF4 $\rightarrow$ Deterministic
- **Intuition:**
  - High-frequency $\rightarrow$ stochastic (noise)
  - Low-frequency $\rightarrow$ deterministic (trend)
- **Note:**
  - Classification is heuristic (no fixed threshold)

ARIMA Model

Model Equation

$$Y_t = \alpha_0 + \sum_{i=1}^{p} \alpha_i Y_{t-i} + \sum_{j=1}^{q} \beta_j \epsilon_{t-j} + \epsilon_t$$

Interpretation

- Y_t : Value today (current value)
- **AR (p):** Depends on past values

$$Y_{t-1}, Y_{t-2}, \ldots$$

- **MA (q):** Depends on past errors

$$\epsilon_{t-1}, \epsilon_{t-2}, \ldots$$

- ϵ_t : White noise (random shock)

Key Idea: Current value depends on past values + past errors

XGBoost (Conceptual Overview)

- XGBoost builds a strong model by combining many simple models
- **Decision Trees**
  - Each tree = simple rules
- **Ensemble Approach**
  - Multiple trees combined together
- **Boosting Mechanism**
  - Trees are built sequentially
  - Each new tree corrects previous mistakes
- **Result**
  - Strong predictive model
  - Captures complex nonlinear patterns

XGBoost: Prediction Model (Eq. 8)

$$y_i = \sum_{k=1}^{K} f_k(x_i)$$

Interpretation:

- y_i : Predicted value
- K : Number of trees
- f_k : Function corresponding to k-th decision tree
- x_i : Input data

Key Idea:

- Prediction = sum of many decision trees

XGBoost: Objective Function (Eq. 9)

$$\text{Obj}(\theta) = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k)$$

Interpretation:

- $l(y_i, \hat{y}_i)$: Prediction error (loss function)
- $\Omega(f_k)$: Complexity penalty for k-th tree
- $\sum \Omega(f_k)$: Total penalty over all trees

Key Idea:

- Minimize error + complexity
- Prevents overfitting

Evaluation Metrics

MAE

$$\text{MAE} = \frac{1}{o}\sum_{l=1}^{o}|Y_l - F_l|$$

RMSE

$$\text{RMSE} = \sqrt{\frac{1}{o}\sum_{l=1}^{o}(Y_l - F_l)^2}$$

MAPE

$$\text{MAPE} = \frac{1}{o}\sum_{l=1}^{o}\left|\frac{Y_l - F_l}{Y_l}\right|$$

MASE

$$\text{MASE} = \frac{1}{n}\sum_{t=1}^{n}\frac{y_t - \hat{y}_t}{\text{MAE}}$$

Diebold–Mariano (DM) Test

Purpose: Compare forecasting models

$$k_l = (y_l - F_{1,l})^2 - (y_l - F_{2,l})^2$$

$$\bar{k} = \frac{1}{o}\sum_{l=1}^{o} k_l$$

$$DM = \frac{\bar{k}}{\sqrt{\mathrm{Var}(\bar{k})}}$$

Interpretation:

- $\bar{k} > 0 \rightarrow$ Model 2 better
- $\bar{k} < 0 \rightarrow$ Model 1 better
- Large $|DM| \rightarrow$ significant difference

Directional Statistic (DS)

Purpose: Predict direction correctly

$$F_l = \begin{cases} 1, & (y_{l+1} - y_l)(\hat{y}_{l+1} - y_l) \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

$$DS = \frac{1}{m} \sum_{l=1}^{m} F_l \times 100$$

Interpretation:

- Measures correct up/down prediction
- Higher DS $\rightarrow$ better model

Paper Data

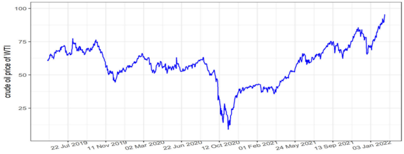
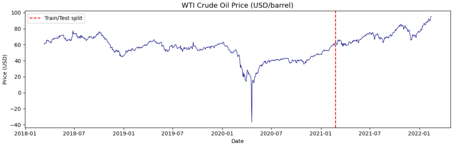


FIGURE 2. WTI market (Dollar per Barrel) daily crude oil prices.

Source: Paper

Our Data (Python)



Train/Test split shown

Paper

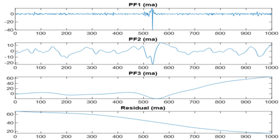
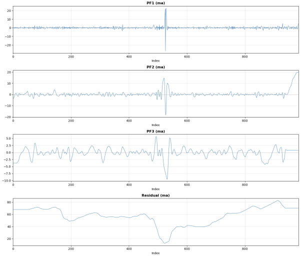


FIGURE 3. Using "LMD" the PFs and residual plots of the WTI crude oil price".

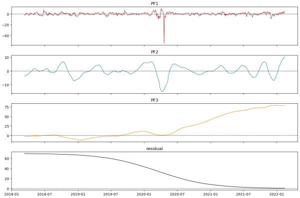
PF1 noisy, PF2–PF3 structured, residual trend

Our Code (Raw)



Similar structure but deviations present

Our Code (Adjusted)



Residual smoothing closer to paper

Average Mutual Information (AMI): Paper vs Theory

Paper (AMI of PFs)

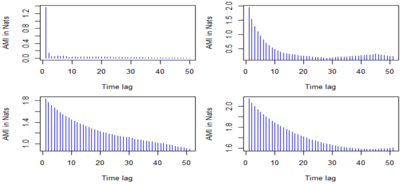
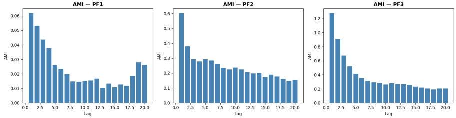


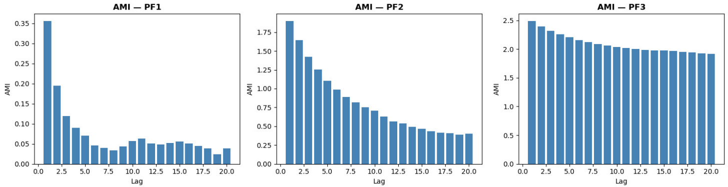
FIGURE 4. "PF's plots for Average Mutual Information (AMI).

Our Computation (Theory)



- PF1 shows rapid decay $\rightarrow$ low dependence (stochastic)
- PF2, PF3 show slower decay $\rightarrow$ structured (deterministic)

Average Mutual Information (AMI): Visual Interpretation



- PF1: Low AMI values → weak dependence → stochastic
- PF2, PF3: Higher AMI → stronger dependence → deterministic
- Confirms classification used in the paper

Prediction Accuracy (WTI): Paper vs Theory

Paper Results

TABLE 3. Accuracy of prediction for "WTI".

Method	MAE	MAPE	RMSE	DS
	A)Five-day-ahead select model results.			
ARIMA	2.02	0.337	2.1973	66.23
XGBOOST	7.4865	0.125	8.0739	69.15
LMD-ARIMA	1.5188	0.0254	1.5917	53.63
LMD-XGBOOST	11.9011	0.1983	13.9352	59.12
LMD-SD-XGBOOST	21.2708	0.3567	21.2954	69.24
LMD-ARIMA-SD-XBOOST	1.3341	0.0241	1.4842	72.34
	B) Ten-day-ahead select model results.			
ARIMA	1.5154	0.026	1.6229	66.67
XGBOOST	8.4678	0.1462	9.7839	69.56
LMD-ARIMA	1.3246	0.0229	1.4936	60.23
LMD-XGBOOST	20.509	0.3543	24.9412	63.12
LMD-SD-XGBOOST	19.879	0.3396	19.9542	69.54
LMD-ARIMA-SD-XGBOOST	1.1587	0.02	1.3881	75.56
	C) Fifteen-day-ahead select model results.			
ARIMA	1.1286	0.0192	1.3483	61.35
XGBOOST	8.6651	0.1482	9.7931	63.35
LMD-ARIMA	1.1785	0.0199	1.4921	65.56
LMD-XGBOOST	19.4858	0.3337	22.8925	70.12
LMD-SD-XGBOOST	19.9708	0.3417	20.0235	71.34
LMD-ARIMA-SD-XGBOOST	0.9464	0.0161	1.1231	76.67

Our Results (Theory)

TABLE 3. Accuracy of prediction for WTI

Method	MAE	MAPE	RMSE	DS
A) Five-day-ahead select model results				
ARIMA	25.4674	31.0643	379.6851	51.02
XGBoost	4.0988	5.1306	6.6210	39.59
LMD-ARIMA	0.8353	1.1639	1.0893	65.71
LMD-XGBoost	3.3353	4.1612	5.5823	58.78
LMD-SD-XGBoost	3.3353	4.1612	5.5823	58.78
LMD-ARIMA-SD-XGBoost	0.8852	1.2349	1.1662	64.49
B) Ten-day-ahead select model results				
ARIMA	25.4674	31.0643	379.6851	51.02
XGBoost	4.0988	5.1306	6.6210	39.59
LMD-ARIMA	0.8415	1.1727	1.0963	64.49
LMD-XGBoost	3.3353	4.1612	5.5823	58.78
LMD-SD-XGBoost	3.3353	4.1612	5.5823	58.78
LMD-ARIMA-SD-XGBoost	0.9033	1.2648	1.2046	62.86
C) Fifteen-day-ahead select model results				
ARIMA	25.4674	31.0643	379.6851	51.02
XGBoost	4.0988	5.1306	6.6210	39.59
LMD-ARIMA	0.8384	1.1685	1.0936	65.71
LMD-XGBoost	3.3353	4.1612	5.5823	58.78
LMD-SD-XGBoost	3.3353	4.1612	5.5823	58.78
LMD-ARIMA-SD-XGBoost	0.9300	1.3028	1.2310	61.22

Prediction Accuracy (WTI): Visual Representation

TABLE 3. Accuracy of prediction for WTI

Method	MAE	MAPE	RMSE	DS
A) Five-day-ahead select model results				
ARIMA	25.4674	31.0643	379.6851	51.02
XGBoost	4.0988	5.1306	6.6210	39.59
LMD-ARIMA	1.1210	1.5785	1.5099	55.51
LMD-XGBoost	3.9152	4.9099	6.1137	50.20
LMD-SD-XGBoost	3.9152	4.9099	6.1137	50.20
LMD-ARIMA-SD-XGBoost	1.1983	1.6860	1.6071	55.10
B) Ten-day-ahead select model results				
ARIMA	25.4674	31.0643	379.6851	51.02
XGBoost	4.0988	5.1306	6.6210	39.59
LMD-ARIMA	1.1192	1.5763	1.5081	55.51
LMD-XGBoost	3.9152	4.9099	6.1137	50.20
LMD-SD-XGBoost	3.9152	4.9099	6.1137	50.20
LMD-ARIMA-SD-XGBoost	1.2176	1.7037	1.5886	51.84

Prediction Accuracy (WTI): Key Insights

Comparison of Models

- Hybrid model (LMD-ARIMA-XGBoost) performs best overall
- Achieves lower error metrics:
  - MAE $\downarrow$
  - MAPE $\downarrow$
  - RMSE $\downarrow$
- Higher Directional Statistic (DS)
  - Better prediction of upward/downward movement
- **Reason:**
  - LMD separates components
  - ARIMA captures linear structure
  - XGBoost captures nonlinear residuals

Forecast Comparison (WTI): First 20 Observations

Our Results (Visual)

TABLE 4. WTI forecasting results (first 20 test observations)

Date	WTI	ARIMA	XGB	LMD-ARIMA	LMD-XGB	LMD-SD-XGB	LMD-ARIMA-SD-XGB
02/24/2021	63.21	61.284	61.235	62.002	62.487	62.487	61.934
02/25/2021	63.43	62.367	63.697	62.987	63.042	63.042	62.815
02/26/2021	61.55	62.966	63.553	63.225	62.960	62.960	63.176
03/01/2021	60.54	62.169	60.081	61.835	61.791	61.791	61.498
03/02/2021	59.70	61.253	61.158	61.358	60.747	60.747	60.621
03/03/2021	61.33	60.379	59.522	60.845	60.313	60.313	59.962
03/04/2021	63.81	60.914	61.278	61.753	61.163	61.163	62.336
03/05/2021	66.08	62.543	63.061	62.952	63.108	63.108	63.431
03/08/2021	65.03	64.539	65.231	64.187	65.245	65.245	66.761
03/09/2021	64.02	64.817	64.337	63.726	64.493	64.493	64.051
03/10/2021	64.45	64.367	64.637	63.196	64.261	64.261	63.544
03/11/2021	66.02	64.414	64.847	63.340	63.088	63.088	64.055
03/12/2021	65.59	65.320	65.785	64.141	64.517	64.517	65.774
03/15/2021	65.36	65.472	65.816	63.857	65.158	65.158	64.829
03/16/2021	64.82	65.409	64.657	63.785	64.948	64.948	63.917
03/17/2021	64.55	65.077	64.054	63.445	63.816	63.816	64.139
03/18/2021	59.95	64.780	65.108	63.272	64.481	64.481	63.226
03/19/2021	61.43	62.053	57.660	60.780	61.239	61.239	58.316
03/22/2021	61.48	61.700	61.068	61.468	60.525	60.525	61.270
03/23/2021	57.75	61.576	61.395	61.506	60.043	60.043	61.688

Our Results (Theory)

TABLE 4. WTI forecasting results (first 20 test observations)

Date	WTI	ARIMA	XGB	LMD-ARIMA	LMD-XGB	LMD-SD-XGB	LMD-ARIMA-SD-XGB
02/24/2021	63.21	61.502	61.235	62.815	63.066	63.066	62.575
02/25/2021	63.43	62.464	63.697	62.947	62.329	62.329	62.746
02/26/2021	61.55	62.858	63.553	62.568	62.329	62.329	62.560
03/01/2021	60.54	61.952	60.081	61.960	61.917	61.917	61.987
03/02/2021	59.70	61.170	61.158	62.448	60.438	60.438	62.332
03/03/2021	61.33	60.512	59.522	62.688	64.756	64.756	64.554
03/04/2021	63.81	61.195	61.278	63.162	62.132	62.132	63.015
03/05/2021	66.08	62.708	63.061	64.003	63.766	63.766	64.086
03/08/2021	65.03	64.393	65.231	65.025	64.708	64.708	62.824
03/09/2021	64.02	64.388	64.337	65.046	64.984	64.984	64.419
03/10/2021	64.45	64.023	64.637	65.091	65.596	65.596	64.737
03/11/2021	66.02	64.445	64.847	65.117	65.239	65.239	64.992
03/12/2021	65.59	65.692	65.785	65.818	65.872	65.872	65.542
03/15/2021	65.36	65.881	65.816	66.262	65.863	65.863	66.389
03/16/2021	64.82	65.662	64.657	65.319	66.037	66.037	65.272
03/17/2021	64.55	65.075	64.054	63.353	64.839	64.839	64.073
03/18/2021	59.95	64.573	65.108	61.568	64.258	64.258	61.720
03/19/2021	61.43	61.755	57.660	60.605	59.505	59.505	60.719
03/22/2021	61.48	61.686	61.068	60.351	60.157	60.157	60.423
03/23/2021	57.75	61.702	61.395	59.506	59.529	59.529	59.122

Forecast Comparison: Key Insights

- Predictions across models are generally close to actual values
- Hybrid model shows more stable and consistent predictions
- ARIMA captures linear trends but misses fluctuations
- XGBoost captures nonlinear patterns but may vary
- Hybrid model combines both $\rightarrow$ improved accuracy

Thank You!